

Yes. **A2 is where we stop talking about what the building means and decide how that meaning becomes physical space.**

We are still **not drawing the final floor plan**. The objective here is to establish the hierarchy strongly enough that when we reach A3/A4, room placement is constrained by logic rather than preference.

A2 — Phantom Q HQ Spatial Hierarchy

The spatial hierarchy should answer:

When somebody enters Phantom Q, what spaces do they encounter first, what sits at the operational centre, what capabilities sit behind secondary boundaries, and what lies deepest inside the facility?

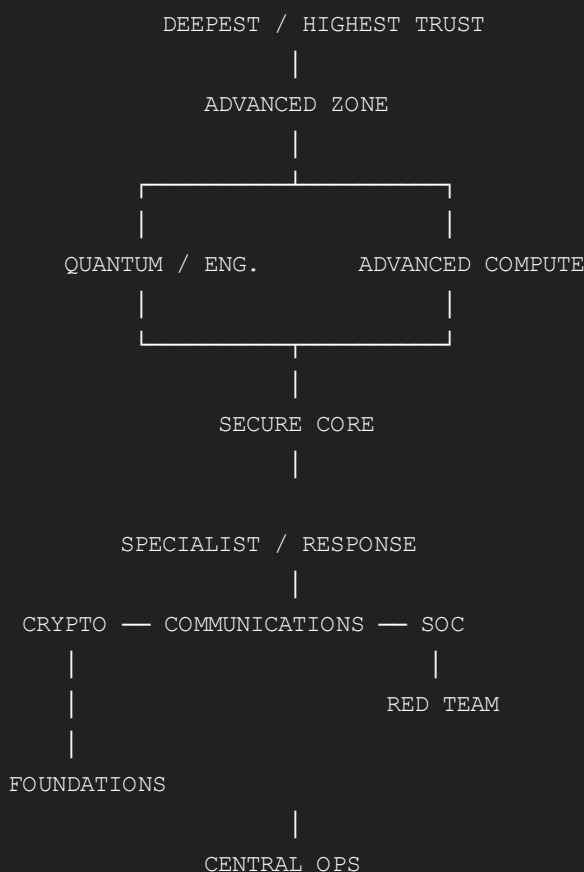
My recommendation is a **hub + controlled wings + deep technical spine**.

Not one long corridor.

Not a random collection of departments.

A2.1 — Overall spatial model

The Headquarters should have four major spatial layers:



RECEPTION / ID

ENTRANCE

This is **hierarchy**, not yet geometry.

The important structure is:

ENTRY

↓

IDENTITY

↓

OPERATIONAL HEART

↓

SPECIALIST CAPABILITY

↓

SECURITY RESPONSE

↓

ADVANCED ENGINEERING

↓

HIGH-TRUST CORE

A2.2 — Zone 1: Arrival and Identity Boundary

This must be the front-most controlled portion of the building.

Contains:

MAIN ENTRANCE

↓

RECEPTION

↓

REGISTRATION / IDENTITY

↓

ACCESS CONTROL GATE

This area exists before the learner is considered fully inside operational Phantom Q.

Security meaning

The boundary asks:

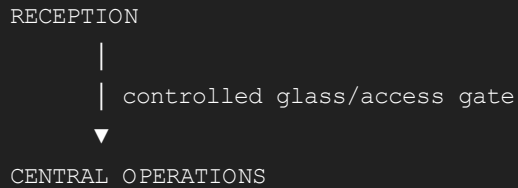
Who are you, and are you permitted into the operational facility?

This directly supports the Prologue.

What should be visible from here?

The player should get a **glimpse of Central Operations**, but not the whole building.

For example:



They should see activity beyond the gate.

That immediately gives the impression:

There is a functioning organisation beyond this point.

A2.3 — Zone 2: Central Operational Core

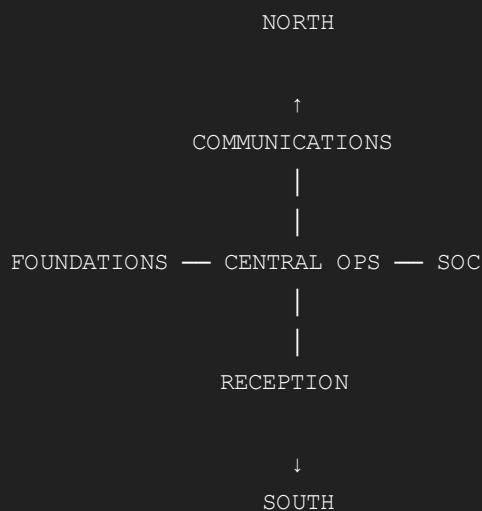
Immediately after the identity boundary should be:

CENTRAL OPERATIONS

This is the spatial anchor of the entire Headquarters.

Everything important should be understandable relative to it.

Conceptually:



Again — not final geometry.

But this establishes four principal directions from the hub.

Why Central Ops belongs directly after Reception

Because the player's progression begins:

```
OUTSIDER
↓
IDENTITY CONFIRMED
↓
OPERATIONAL TRAINEE
```

The first environment they are trusted to work in should therefore be the **normal operational centre**.

That's where Workstation 04 belongs.

A2.4 — Central Operations needs to function as a navigation landmark

From Central Operations, the learner should be able to orient themselves quickly.

Ideally, they can visually distinguish:

```
FOUNDATIONS ROUTE
CRYPTO / COMMUNICATIONS ROUTE
SOC ROUTE
ADVANCED TECHNICAL ROUTE
```

without opening a minimap.

This can be achieved through:

- sightlines;
 - signage;
 - colour/material differences;
 - glass partitions;
 - distinct door architecture;
 - lighting;
 - large departmental identifiers.
-

A2.5 — Zone 3A: Foundations should be close to the Core

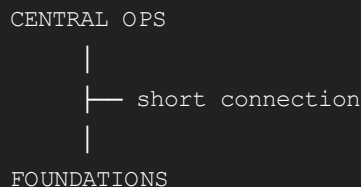
Training & Foundations should be one of the **closest specialist spaces to Central Operations**.

Why?

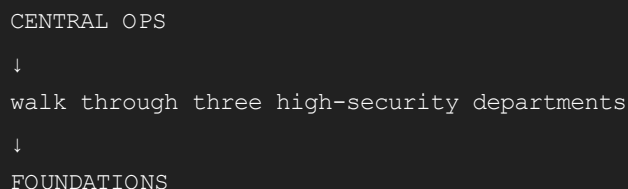
Because it is:

- an early-career capability;
- frequently accessible;
- lower sensitivity than SOC/Red Team;
- the first newly unlocked space after the Prologue.

So spatially:



not:



That would make no organisational sense.

A2.6 — Foundations should be visible very early

During the Prologue the learner should potentially see something such as:

```
graph TD; S[TRAINING & FOUNDATIONS]; S --> A[ACCESS:]; A --> R[RESTRICTED - TRAINEE ASSESSMENT REQUIRED];
```

A diagram showing a sign with the text 'TRAINING & FOUNDATIONS' at the top. Below it, the text 'ACCESS:' is followed by 'RESTRICTED - TRAINEE ASSESSMENT REQUIRED'.

Then after PQ-001:

ACCESS GRANTED

The same physical door changes state.

That is our progression principle embodied spatially.

A2.7 — Zone 3B: Cryptography and Communications should form a linked specialist cluster

This relationship is important enough that I would deliberately design around it.

CRYPTOGRAPHY
↓
COMMUNICATIONS

Because:

CRYPTO
How is information protected?

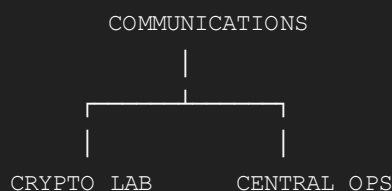
COMMUNICATIONS
How does information move?

The player needs to travel between them repeatedly in Incidents 02–05.

Therefore these spaces should be **near one another**.

Recommended hierarchy

Something like:



or:

CENTRAL OPS
|
COMMUNICATIONS

A3 will determine the exact adjacency.

But the hierarchy rule should be:

Crypto and Communications belong to the same operational-specialist cluster.

A2.8 — Communications should be more central than Crypto

This is an important subtlety.

Communications supports normal operations, SOC investigation and cryptographic missions.

Therefore Communications needs relationships with:

CENTRAL OPS
CRYPTO
SOC

That makes Communications something of a **bridge department**.

Conceptually:

CRYPTO
 \
 \
CENTRAL OPS — COMMUNICATIONS — SOC

This may be one of the most useful organising relationships in the entire building.

It gives us:

NORMAL OPERATIONS
 ↓
COMMUNICATIONS
 ↓
SECURITY MONITORING

while Crypto can influence the protected communication process from the other side.

A2.9 — Zone 4A: SOC should sit close to Communications and Central Ops

SOC should not be buried at the deepest point of the HQ.

Security Operations needs rapid access to:

- operational systems;
- communications;
- incident data;
- command personnel.

So:

CENTRAL OPS



SOC

should be a relatively short relationship.

And:

COMMUNICATIONS



SOC

should be even stronger.

That matches Incident 04 extremely well:

NORMAL OPERATION



COMMUNICATION PROBLEM



SOC ESCALATION

The spatial journey reflects the incident escalation.

A2.10 — SOC must still have a stronger access boundary

Even though it is physically near the Core, it should be **security-separated**.

Think:

CENTRAL OPS

|

| secure glass partition

|

| access-controlled door

▼

SOC

The player can see SOC staff activity earlier.

But cannot simply wander inside.

This is exactly the type of **visible future capability** we wanted.

A2.11 — Zone 4B: Red Team should sit behind SOC, not beside Reception

This one I would strongly recommend.

The operational relationship is:

SOC

detects / investigates

↓

RED TEAM

actively engages under authority

Therefore Red Team should feel **one security layer deeper than SOC**.

Conceptually:

CENTRAL OPS

|

▼

SOC

|

SECONDARY SECURITY GATE

|

▼

RED TEAM OPERATIONS

Not necessarily literally in one straight corridor, but in trust hierarchy.

Why this is good for the campaign

During earlier incidents:

The learner might enter SOC or see Red Team behind another boundary:

RED TEAM OPERATIONS

CLEARANCE REQUIRED

Incident 05:

RED TEAM

STANDBY

Incident 06:

MISSION AUTHORISATION

ACCESS GRANTED

That makes the player's progression physically satisfying.

A2.12 — Red Team should have controlled connection to HQ authority

One thing we need to remember from Incident 06:

Red Team **cannot self-authorise offensive actions**.

Therefore the architecture should not imply that Red Team is an independent uncontrolled bunker.

There needs to be a believable operational relationship:

RED TEAM

↑

SOC

↑

CENTRAL/HQ COMMAND

The actual signed authority is digital, but spatially the organisation should still make sense.

A2.13 — Zone 5: Engineering should form the gateway to the advanced technical wing

Engineering is where I would start the **deep technical hierarchy**.

Why Engineering first?

Because it connects the ordinary cyber/digital world to:

- embedded hardware;
- electronics;
- physical devices;
- QKD hardware;
- instrumentation.

So rather than placing Quantum Lab randomly at the back, use:

```
OPERATIONAL HQ
    ↓
ENGINEERING
    ↓
ADVANCED TECHNICAL CAPABILITY
```

This creates a believable transition.

A2.14 — Engineering should be accessible without passing through Quantum

Hierarchy:

```
ENGINEERING
  lower specialist restriction

QUANTUM
  higher specialist restriction
```

not the other way around.

That allows future non-quantum engineering gameplay too.

A2.15 — Quantum Wing should be deeper than general Engineering

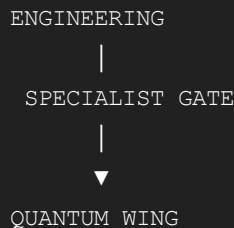
The Quantum environment contains:

- optical/QKD demonstrators;

- specialist hardware;
- Alice/Bob/Eve stations;
- potentially lasers/optics;
- measurement equipment;
- advanced experimental infrastructure.

Therefore it deserves another security boundary.

Conceptually:



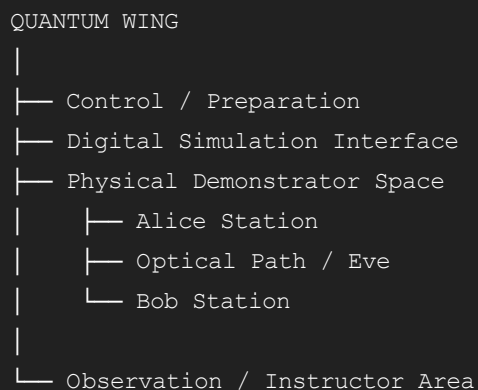
This makes the physical QKD demonstrator feel like serious infrastructure.

A2.16 — Physical QKD requires a larger spatial node

Quantum Wing should not simply be another small office.

Hierarchy should reserve space for a substantial lab.

Conceptually:



We can define exact rooms later.

But A2 must acknowledge that this wing carries **physical depth and equipment footprint**.

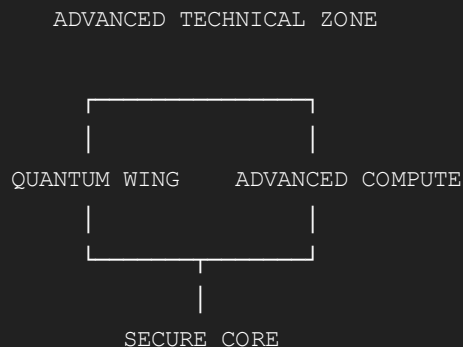
A2.17 — Advanced Compute should be parallel to Quantum, not necessarily inside it

I would avoid assuming:

```
Quantum
↓
Advanced Compute
```

as if one is always subordinate to the other.

Better:



This gives Advanced Compute independent future value.

It could later support:

- quantum-threat simulation;
- large computational tasks;
- attack modelling;
- advanced security simulation;
- digital twins.

A2.18 — Secure Core must be the deepest point

This is the highest-trust location.

It should not be on the player's normal travel route.

The architecture should communicate:

```
YOU DO NOT PASS THROUGH
THE SECURE CORE
TO GET SOMEWHERE ELSE.
```

It is a destination.

Not a corridor.

Its hierarchy:

```
GENERAL HQ
↓
SPECIALIST
↓
ADVANCED
↓
SECURE CORE
```

This creates a powerful future campaign location.

A2.19 — The Secure Core should be visually teased, not explained early

From some advanced-area viewpoint, perhaps the learner eventually sees:

```
SECURE CORE

ACCESS LEVEL:
RESTRICTED
```

But early in Release 1 they do not need an explanation of everything behind it.

Mystery is useful.

A2.20 — Proposed hierarchy bands

We can now convert the structure into explicit bands.

Spatial band	Areas	Meaning
Band 0 — External	Exterior/Approach	Not yet inside Phantom Q
Band 1 — Controlled Entry	Reception, Identity, Access Control	Establish who may enter
Band 2 — Operational Core	Central Operations, Communications	Normal daily operation

Spatial band	Areas	Meaning
Band 3 — Specialist	Foundations, Crypto, Engineering	Skill/capability development
Band 4 — Security Response	SOC	Monitoring/investigation
Band 5 — Active Security	Red Team	Authorised offensive response
Band 6 — Advanced Technical	Quantum, Advanced Compute	Advanced specialist capability
Band 7 — High Trust	Secure Core	Highest-restriction capability

These aren't necessarily player levels.

They are **architectural security-depth bands**.

A2.21 — Not all hierarchy means physical distance

Important.

For example:

SOC can be physically only 8 metres from Central Ops but still be a higher-security zone because there is:

```
glass
access control
restricted door
monitoring
```

So hierarchy is created by:

```
DISTANCE
+
DOORS
+
VISIBILITY
+
ACCESS CONTROL
+
ARCHITECTURAL LANGUAGE
```

not just by making every advanced room very far away.

This lets us maintain a compact Headquarters.

A2.22 — Main circulation hierarchy

We should have three types of circulation.

Primary circulation

The routes a basic trainee uses frequently.

Entrance
Reception
Central Ops
Foundations
basic Communications access

These should be broad and easy to navigate.

Approx:

3.0 m corridors

Secondary specialist circulation

For:

Crypto
SOC
Engineering

Still comfortable, but increasingly controlled.

Approx:

2.4–3.0 m

Restricted circulation

For:

Red Team
Quantum
Advanced Compute
Secure Core

These can be more enclosed and security-heavy.

But still never become a maze.

Approx:

2.4 m minimum

A2.23 — We should avoid one giant main hallway

Something like this would be poor:

ENTRANCE

Reception
Central Ops
Training
Crypto
Communications
SOC
Red Team
Engineering
Quantum
Secure Core

It would feel like a hospital or school corridor.

We want:

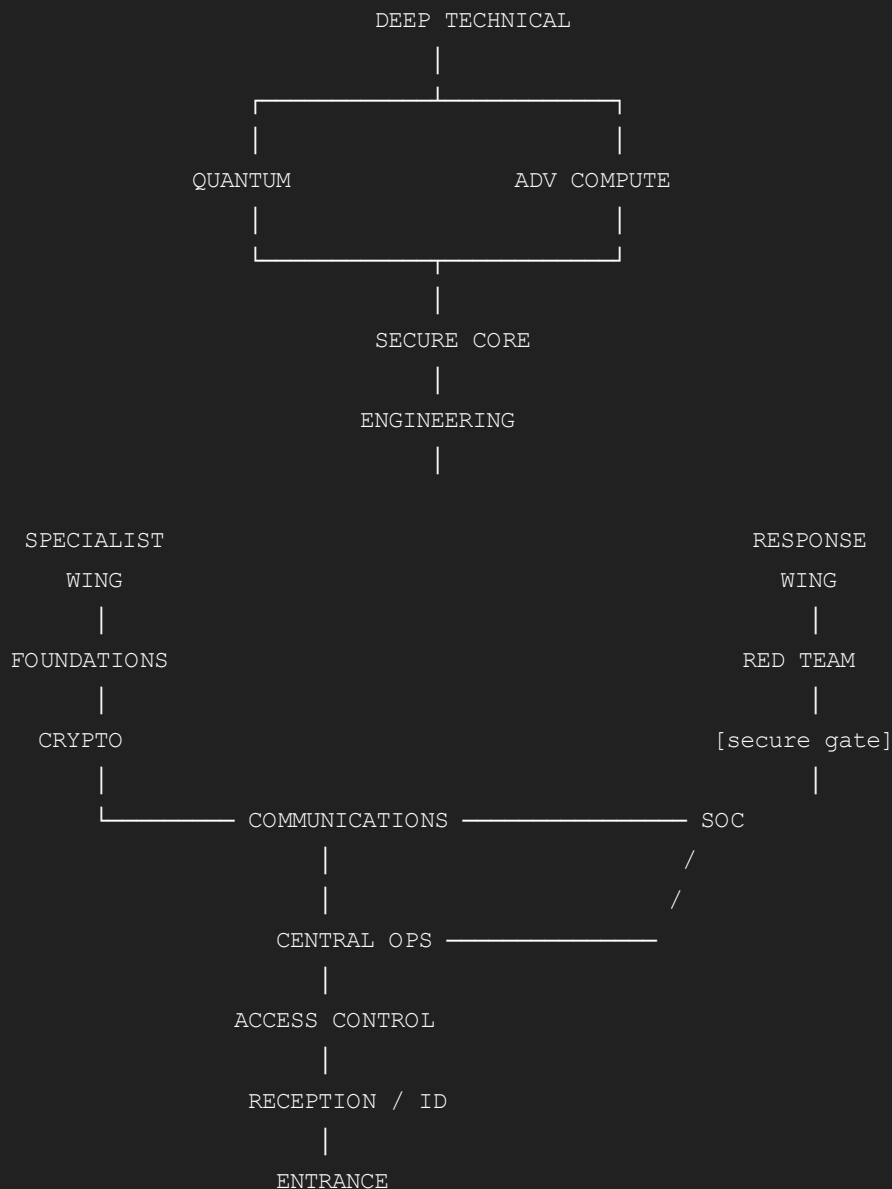
hub and branches

because that creates:

- orientation;
- departmental identity;
- meaningful access thresholds.

A2.24 — Proposed conceptual topology

I think our hierarchy is converging toward something like this:



Do not lock the shape.

Lock the **relationship logic**.

A2.25 — Campaign test against this hierarchy

Now loop back through our incidents.

Prologue

```
Entrance
↓
Reception / ID
```

Excellent.

Minimal movement.

Incident 01

Central Ops

↓

Foundations

Short route.

Makes sense for first unlock.

Incident 02

Central Ops

↓

Crypto

↔

Communications

Both are logically clustered.

Good.

Incident 03

Crypto

↔

Communications

Very little unnecessary travel.

Excellent.

Incident 04

```
Central Ops
↓
Communications
↓
SOC
```

Excellent escalation path.

Incident 05

```
SOC
↔
Communications
↔
Crypto
↓
Red Team standby
```

These are all relatively nearby around the operational/security side of HQ.

Good.

Incident 06

```
SOC
↔
Red Team
↔
HQ command
```

Again compact and operational.

Then:

```
restore
↓
Central Ops
```

Good spatial closure.

So A2 currently supports the campaign very well.

A2.26 — Character-flow test

Alice

Her primary environments:

```
Central Ops  
Crypto  
Communications
```

All relatively close.

Good.

Eve

Primary likely environments:

```
Reception/Security  
Communications  
SOC
```

Again logically connected.

Crucially, she doesn't need to wander through Red Team/Quantum for basic security support.

Good.

Bob

Mostly remote.

His “presence” can come through:

```
Central Ops terminal  
Communications  
Crypto interfaces
```

No physical-path problem.

Red Team

SOC

↔

Red Team Operations

Direct operational relationship.

Good.

Engineering/QKD characters later

Engineering

↔

Quantum

Natural relationship.

Good.

So A2 also passes the character test.

A2.27 — Camera test conceptually

The hub + short-wing approach is very compatible with our camera.

Why?

Because we can frame:

CENTRAL OPS

as a large open landmark.

Wings can extend outward without creating extremely deep sightlines.

The player enters a wing:

camera follows

↓

upper walls cut away

↓

room becomes primary visual field

Then returns to the hub.

Much easier than navigating a huge dense rectangular office building.

A2.28 — Handling test

This topology should keep most routine travel under roughly:

10–25 seconds

for early operational areas.

Deeper advanced spaces can take slightly longer because that journey should feel meaningful.

But we should avoid:

```
Incident 05:  
walk 90 seconds to Crypto  
walk 90 seconds back  
walk 90 seconds to SOC
```

That's not educational gameplay.

The hierarchy must stay compact.

A2.29 — Visual distinction should increase with depth

We should not use random themes, but there should be progression.

Entry

```
clean  
welcoming  
controlled  
professional
```

Central Operations

```
active  
open  
information-rich
```

Specialist

technical
equipment
workbenches

SOC

monitoring
dense operational displays
lower ambient lighting perhaps

Red Team

focused
restricted
mission-orientated

Engineering

hardware
test equipment
fabrication/integration

Quantum

precision
optics
controlled laboratory environment

Secure Core

minimal
high-security
deliberately controlled

Still one Phantom Q visual family.

A2.30 — What A2 should now lock

I think we can safely carry the following into A3:

Spatial hierarchy lock

1. Entrance → Reception/Identity → Access Control → Central Operations is the primary arrival axis.
2. Central Operations is the main spatial hub.
3. Foundations is an early-access specialist space close to Central Ops.
4. Crypto and Communications form a functional cluster.
5. Communications is a bridge between Central Ops, Crypto and SOC.
6. SOC is close to normal operations but behind a stronger security boundary.
7. Red Team is one security layer deeper than SOC and requires separate authorisation.
8. Engineering begins the deeper technical portion of HQ.
9. Quantum Wing is deeper/more specialist than general Engineering.
10. Advanced Compute is a parallel advanced capability to Quantum, rather than merely a sub-room of it.
11. Secure Core is the deepest/highest-trust destination and is never used as a passageway.
12. The HQ uses hub + short wings, not a single long corridor or maze.
13. Operational travel between Prologue–Incident 06 areas remains short.
14. Restricted areas can often be visible before they are accessible.
15. Spatial depth communicates increasing trust/capability without relying only on distance.
16. The hierarchy already reserves meaningful space for physical QKD.

And importantly, after looping A1 ↔ A2, I do not see a contradiction yet.

The architecture is becoming coherent:

```
IDENTITY
↓
OPERATIONS
↓
SPECIALISATION
↓
INVESTIGATION
↓
AUTHORISED RESPONSE
↓
ADVANCED ENGINEERING
↓
HIGH TRUST
```

The next step, **A3 — Department Adjacency**, is where we become much more exact. Instead of merely saying “Crypto should be near Communications”, we will establish a relationship matrix of **MUST directly connect** / **SHOULD be near** / **CAN be separated** /

SHOULD NOT directly connect for every department. That will heavily constrain the final floor plan and prevent arbitrary room placement.